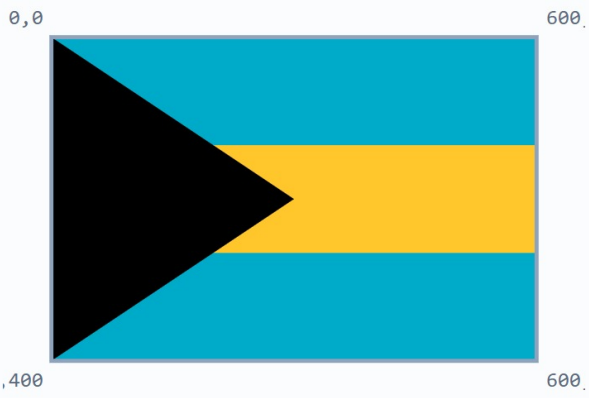
Draw a Flag



The Bahamas · canvas 600 × 400

What you are making

Three equal horizontal bands — aquamarine, gold, aquamarine — with a black triangle pointing in from the left edge.

Colors:

- aquamarine `fill(0, 171, 201)`
- gold `fill(255, 199, 44)`
- black `fill(0)`

> The three bands are **equal thirds** of the height. Use `height/3`, not `133`.

Set up your sketch

1. In the p5.js editor: `File → New`
2. Name it `u1l02-flag` and **save**
3. Type the code below — do not paste it from anywhere

Start here

```
function setup() {
  createCanvas(600, 400);
}

function draw() {
  noStroke();

  // the three bands
  fill(0, 171, 201);
  rect(0, 0, width, height / 3);

  // TODO: gold band, the middle third

  // TODO: aquamarine band, the bottom third

  // TODO: the black triangle
}
```

> **Why no `background()`?** The three bands cover the whole canvas between them, so nothing is left to show through. If you see white anywhere, a band is the wrong size.

The triangle

Three corners, and every one of them can be written using `width` and `height`:

- the **top-left corner** of the canvas
- the **bottom-left corner** of the canvas
- a point on the left edge... no — the tip reaches **halfway across**, level with the middle

> **Work the tip out before you type it.** Halfway across is `width/2`. Level with the middle is `height/2`. If you typed `300` and `200`, it looks right — until the next paragraph.

Check it

1. Does it match the picture?
2. Change `createCanvas(600, 400)` to `createCanvas(900, 600)` and run it again. **The flag should look exactly the same, just bigger.**
3. If any part is now in the wrong place, you typed a number where you should have computed one. Fix it and test again.

> Step 2 is the real test. A flag built from `width` and `height` survives a resize. A flag built from `300` and `200` falls apart, and it is the only way to tell the difference.

Turning it in

1. **Save** your sketch.
2. `File → Share`
3. Copy the **View-Only** link. Not Edit, not Embed.
4. Paste it as a **comment** on the Classroom post for this activity.
5. Open your own link in a new tab and check it works.

> **A sketch you never saved shares an empty link.** Save first, then share. Step 5 takes ten seconds and catches it.

Finished early?

- Draw the flag of **Czechia** instead — two bands and a triangle, and the tip only reaches halfway.
- Or **Kuwait** — three bands and a black trapezoid, which you would have to build from a rectangle and a triangle.
- Or your own: pick any flag that is made of rectangles and triangles, and build it so it survives a resize.

Submit the link even if you are not finished. The link points at your sketch, not a snapshot, so anything you fix later shows up on the same link. This is practice for the rocket you build next lesson — if your flag survives the resize check, you are ready for it.